Auteur: Jeremi Lorek
Studierichting: Informatica
Oefeningenreeks 2E nr. 26.10

$$\log_3(x-3) = \frac{1}{2\log_2 3} + \log_{81}(3x-13)^2$$
$$\Rightarrow \log_3(x-3) = \frac{\log_3 2}{2} + \frac{\log_3(3x-13)^2}{\log_3 81}$$
$$\Rightarrow \log_3(x-3) = \frac{\log_3 2}{2} + \frac{2\log_3(3x-13)}{4}$$
$$\Rightarrow \log_3(x-3) = \frac{\log_3 2}{2} + \frac{\log_3(3x-13)}{2}$$
$$\Rightarrow \log_3(x-3) = \frac{\log_3 2 + \log_3(3x-13)}{2}$$
$$\Rightarrow \log_3(x-3) = \frac{\log_3\left(2(3x-13)\right)}{2}$$
$$\Rightarrow 2\log_3(x-3) = \log_3(6x-26)$$
$$\Rightarrow \log_3(x-3)^2 = \log_3(6x-26)$$
$$\Rightarrow (x-3)^2 = 6x-26$$
$$\Rightarrow x^2 - 6x + 9 = 6x - 26$$
$$\Rightarrow x^2 - 12x + 35 = 0$$
$$\Delta = 144 - 4 \cdot 35 = 4$$
$$x_{1,2} = \frac{12 \pm 2}{2}$$
$$x_1 = 7, \quad x_2 = 5$$

References